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Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 2

1. A production possibilities frontier:

- ☒ A. shows the maximum attainable combinations of two goods that may be produced with available resources.
- ☐ B. shows how unlimited wants exceed the limited resources available to fulfill those wants.
- ☐ C. shows how participants in the market are linked.
- ☐ D. shows the market for a good or service.
- ☐ E. shows the act of buying and selling.

We can show economic efficiency:

- ☐ A. with points on and outside the production possibilities frontier.
- ☐ B. with points inside the production possibilities frontier.
- ☐ C. with points outside the production possibilities frontier.
- ☐ D. with points inside and on the production possibilities frontier.
- ☒ E. with points on the production possibilities frontier.

We can show economic inefficiency:

- ☐ A. with points inside and on the production possibilities frontier.
- ☐ B. with points on the production possibilities frontier.
- ☐ C. with points outside the production possibilities frontier.
- ☐ D. with points on and outside the production possibilities frontier.
- ☒ E. with points inside the production possibilities frontier.

The production possibilities frontier will shift outward

- ☐ A. if resources are used to produce consumption goods.
 - ☒ B. if technological advances occur.
 - ☐ C. if production occurs outside the production possibilities frontier.
 - ☐ D. if resources are not used in production.
 - ☐ E. if resources are not used to produce capital goods.
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2.

You have exams in economics and chemistry coming up and five hours available for studying. The following table shows the trade-offs you face in allocating the time you will spend in studying each subject.

Hours Spent Studying			Midterm Score	
Choice	Economics	Chemistry	Economics	Chemistry
A	5	0	94	73
B	4	1	92	80
C	3	2	89	86
D	2	3	85	91
E	1	4	80	93
F	0	5	74	94

a. Use the data in the table to draw a production possibilities frontier¹ graph.

1.) Use the multipoint curve drawing tool to plot the production possibilities frontier curve. Plot points for all choices A through F. Properly label this curve.

2.) Use the point drawing tool to plot all choices A through F. Properly label these points.

Carefully follow the instructions above, and only draw the required objects.

b. If you are at choice D, the opportunity cost² of increasing your chemistry score by studying one additional hour for the chemistry exam is fewer point(s) on your economics exam. (Enter your response as a whole number.)

c. Under what circumstances would choice A be a sensible choice? Select all that apply.

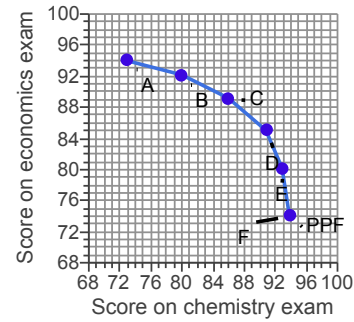
- ☒ A. The economics midterm accounts for a large portion of your economics grade and the chemistry midterm is a small portion of your chemistry grade.
- ☐ B. You are majoring in chemistry and do not require economics to graduate.
- ☒ C. You are majoring in economics and do not require chemistry to graduate.
- ☐ D. The chemistry midterm accounts for a large portion of your chemistry grade, and the economics midterm is a small portion of your economics grade.
- ☐ E. You have a double major in economics and chemistry and require decent grades in both courses to graduate.

1: Definition

A production possibilities frontier (PPF) is a curve showing the maximum attainable combinations of two products that may be produced with available resources and current technology.

2: Definition

The opportunity cost of any activity is the highest-valued alternative that must be given up to engage in that activity.



After plotting the final point of your multipoint curve, press the Esc key on your keyboard to end the line.

3. Suppose your provincial government is deciding which of two sports programs it will pay for (assuming that only one program will be funded). The choices are Sport A, which will allow 24 students to play for 8 months and costs \$25,000 per year, and Sport B, which will allow 36 students to play for 8 months and costs 38,000 per year.

What factors should the provincial government take into account in making this decision of which sports programs it will pay for?

- ☐ A. The funds available for sports programs.
- ☐ B. The number of students, which can play in each sports program.
- ☐ C. The duration of each sports program.
- ☒ D. The relative cost of each sports program.

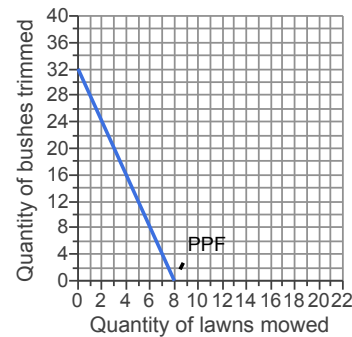
Which sports program would you recommend? Sport A .

4. Tom's lawn service specializes in mowing lawns and trimming bushes. Tom has **4 hours** to devote to lawn services. In 1 hour, he can mow 2 lawns or trim 8 bushes.

Use the line drawing tool to graph Tom's production possibilities frontier given he has 4 hours to devote to lawn services. Properly label this line.

Carefully follow the instructions above, and only draw the required objects.

Tom's opportunity cost of mowing lawns in terms of trimming bushes is trimmed bushes per mowed lawn. (Enter your response rounded to two decimal places.)

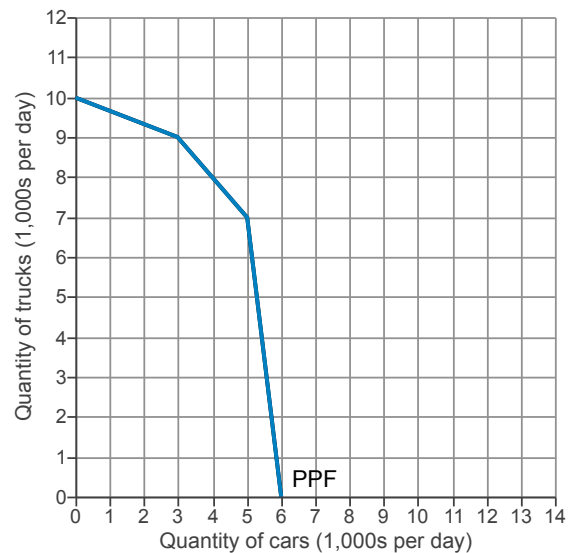


5. The figure to the right illustrates the trade-offs facing Ford Motor Company. The line in the graph is Ford's production possibilities frontier.

If Ford uses all its resources to produce trucks, how many can it produce? thousand trucks per day. (Enter a numeric response using a real number rounded to one decimal place.)

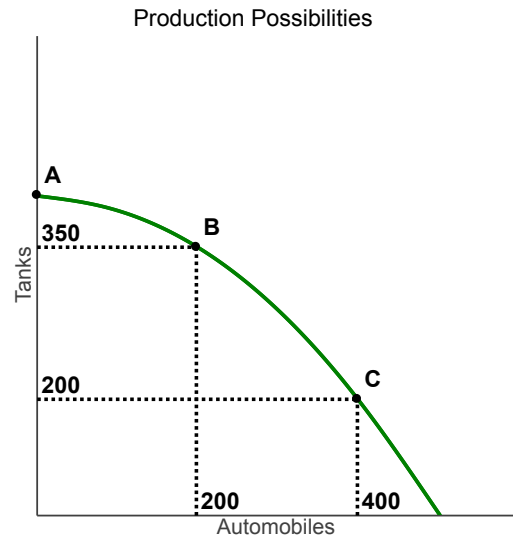
Suppose Ford is currently building 3 thousand cars per day. To build an additional 1 thousand cars, how many fewer trucks can be built? thousand fewer trucks per day.

According to the graph, building 3 thousand cars per day and 3 thousand trucks per day is technically inefficient .



6. On the diagram to the right, movement along the curve from points A to B to C illustrates

- ☒ A. increasing marginal opportunity costs.
- ☐ B. reflexive marginal opportunity costs.
- ☐ C. constant marginal opportunity costs.
- ☐ D. decreasing marginal opportunity costs.



7. What is absolute advantage?

- ☐ A. The gain from selling a product for more than it costs to produce that product.
- ☐ B. The ability to produce a good or service at a lower opportunity cost than other producers.
- ☒ C. The ability to produce more of a good or service than competitors using the same amount of resources.
- ☐ D. The gain from consuming a product whose benefit is greater than its cost.
- ☐ E. The ability to use all available resources to produce output.

What is comparative advantage?

- ☐ A. The ability to use all available resources to produce output.
- ☐ B. The gain from selling a product for more than it costs to produce that product.
- ☒ C. The ability to produce a good or service at a lower opportunity cost than other producers.
- ☐ D. The ability to produce more of a good or service than competitors using the same amount of resources.
- ☐ E. The gain from consuming a product whose benefit is greater than its cost.

Is it possible for a country to have a comparative advantage in producing a good without also having an absolute advantage? A country without an absolute advantage in producing a good

- ☐ A. will have a comparative advantage if it devotes more resources toward that good's production.
- ☒ B. will have a comparative advantage if it has a lower opportunity cost of producing that good.
- ☐ C. will not have a comparative advantage because it has fewer resources.
- ☐ D. will have a comparative advantage if it is able to produce that good at a low total cost.
- ☐ E. will have a comparative advantage if it produces more efficiently.

8. What is the basis for trade?

- ☒ A. Comparative advantage.
- ☐ B. Available resources.
- ☐ C. Efficiency.
- ☐ D. Economic growth.
- ☐ E. Absolute advantage.

How can a country gain from specialization and trade?

- ☐ A. A country can specialize by using all available resources to invest in capital goods to promote economic growth.
 - ☐ B. A country can specialize in producing that which is most scarce and then trade for other needed goods and services.
 - ☒ C. A country can specialize in producing that for which it has a comparative advantage and then trade for other needed goods and services.
 - ☐ D. A country can specialize by using all available resources to produce goods and services to avoid trading with other countries.
 - ☐ E. A country can specialize in producing that for which it has an absolute advantage and then trade for other needed goods and services.
-

9. The graph to the right shows how many kilograms of apples and kilograms of cherries you and your neighbour can each pick in one week. For example, if you devote all of your time to picking apples and none of your time to picking cherries, you can pick 12 kilograms of apples. If you devote all of your time to picking cherries, you can pick 72 kilograms. At the same time, if your neighbour devotes all of her time to picking apples, she can pick 32 kilograms of apples. If she devotes all of her time to picking cherries, she can pick 32 kilograms.

Suppose initially that you (Y) are consuming 8 kilograms of apples and 24 kilograms of cherries and that your neighbour (N) is consuming 4 kilograms of apples and 28 kilograms of cherries, as indicated in the graph. Then, suppose you and your neighbour specialize by each only picking the good for which you have a comparative advantage³ and trade⁴. In particular, suppose you trade your neighbour half of your production for half of what your neighbour produces.

In the table below, first fill in production when specializing.
(Enter numeric responses using integers.)

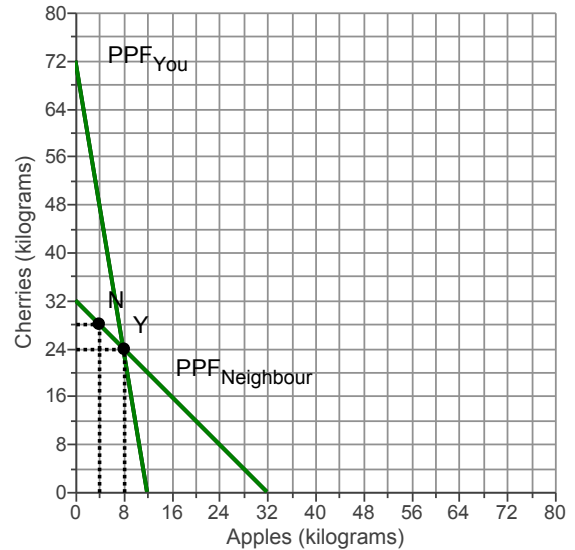
	You		Your Neighbour	
	Apples	Cherries	Apples	Cherries
Consumption without trade	8	24	4	28
Production with Trade	0	72	32	0

Next, fill in consumption with trade. Recall that you trade your neighbour half of what you pick for half of what your neighbour picks.

	You		Your Neighbour	
	Apples	Cherries	Apples	Cherries
Consumption without trade	8	24	4	28
Production with Trade	0	72	32	0
Consumption with Trade	16	36	16	32

Finally, fill in gains from trade.

	You		Your Neighbour	
	Apples	Cherries	Apples	Cherries
Consumption without trade	8	24	4	28
Production with Trade	0	72	32	0
Consumption with Trade	16	36	16	32
Gains from Trade	8	12	12	8



3: Definition

Comparative advantage is the ability of an individual, a firm, or a country to produce a good or service at a lower opportunity cost than competitors.

4: Definition

Trade is the act of buying or selling.

10. Suppose that France and Germany both produce wine and schnitzel. The table below shows combinations of the goods that each country can produce in a day.

France		Germany	
Wine (Bottles)	Schnitzel (Kilograms)	Wine (Bottles)	Schnitzel (Kilograms)
0	12	0	20
1	9	1	16
2	6	2	12
3	3	3	8
4	0	4	4
		5	0

a. Who has the comparative advantage⁵ in producing wine, and who has the comparative advantage in producing schnitzel?

- ☒ A. France has a comparative advantage producing wine, and Germany has a comparative advantage producing schnitzel.
- ☐ B. Germany has a comparative advantage producing wine and schnitzel.
- ☐ C. France has a comparative advantage producing wine and schnitzel.
- ☐ D. France has a comparative advantage producing schnitzel, and Germany has a comparative advantage producing wine.
- ☐ E. Neither has a comparative advantage producing wine or schnitzel.

b. Suppose that France is currently producing 1 bottle of wine and 9 kilograms of schnitzel and Germany is currently producing 3 bottles of wine and 8 kilograms of schnitzel. Then, assume instead that France and Germany specialize by producing only the good for which they have a comparative advantage and then trade⁶ 3 bottles of wine for 11 kilograms of schnitzel.

After specialization and trade, France gains by consuming the same amount of wine and additional kilogram(s) of schnitzel.

(Enter your response using an integer.)

After specialization and trade, Germany gains by consuming the same amount of wine and additional kilogram(s) of schnitzel.

(Enter your response using an integer.)

5: Definition

Comparative advantage is the ability of an individual, a firm, or a country to produce a good or service at a lower opportunity cost than competitors.

6: Definition

Trade is the act of buying or selling.

11. Whether carried out by an individual or a country, production beyond the production possibilities frontier

- ☐ A. can occur by acquiring more productive resources.
- ☐ B. happens as a result of forced saving.
- ☒ C. is not physically possible.
- ☐ D. is possible only through trade.

With respect to consumption, individuals and countries

- ☐ A. are constrained to consume on or inside their production possibilities frontiers.
- ☐ B. can consume beyond their production possibilities frontiers only during periods of economic prosperity.
- ☒ C. can, through trade, consume beyond their production possibilities frontiers.
- ☐ D. can, by choosing not to save for the future, consume beyond their production possibilities frontiers.

12. Suppose the United States and Mexico both produce hamburgers and tacos. The combinations of the two goods that each country can produce in one day are presented in the table below.

United States		Mexico	
Hamburgers (in tons)	Tacos (in tons)	Hamburgers (in tons)	Tacos (in tons)
0	108	0	60
40	72	8	40
80	36	16	20
120	0	24	0

Which country has an absolute advantage in producing tacos? The United States .

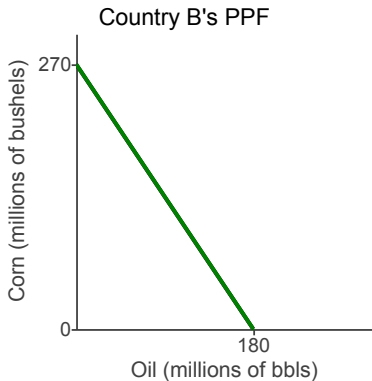
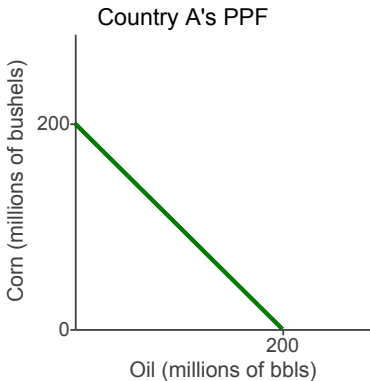
Which country has a comparative advantage in producing tacos? Mexico .

Suppose the United States is currently producing 80 tons of hamburgers and 36 tons of tacos and Mexico is currently producing 16 tons of hamburgers and 20 tons of tacos. If the United States and Mexico each specialize in producing only one good (the good for which each has a comparative advantage), then a total of additional ton(s) of hamburgers can be produced for the two countries combined (*enter a numeric response using an integer*) and a total of additional ton(s) of tacos can be produced.

13. If country ABC can produce a unit of good 1 with fewer resources than can country XYZ, it is correct to say that country ABC

- ☐ A. will not wish to trade good 1 with country XYZ.
- ☒ B. has an absolute advantage in producing good 1.
- ☐ C. has a comparative advantage in producing good 1.
- ☐ D. will export good 1.

14.



From the PPFs shown above it can be determined that the opportunity cost of oil is lower in country A .

Given the comparative opportunity costs as revealed by the PPFs shown above, the comparative advantage for country B lies in corn .

After these two countries specialize and trade with each other, country A will be importing corn .

15.

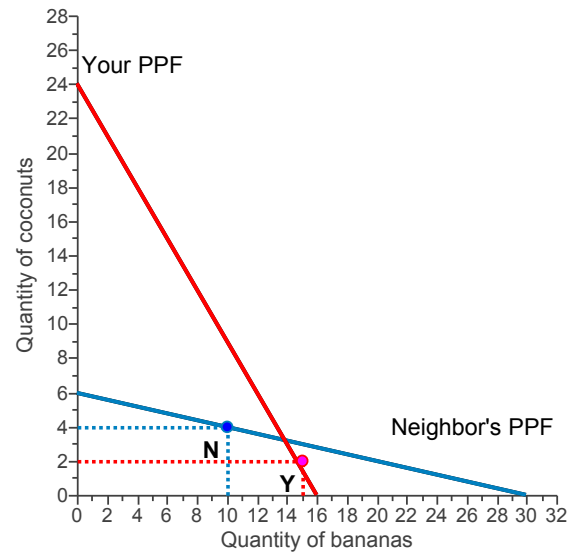
The production possibilities frontiers in the figure to the right show how many bananas and coconuts you (Y) and your neighbor (N) can consume without trade. Suppose you are initially consuming 15 bananas and 2 coconuts and your neighbor is initially consuming 10 bananas and 4 coconuts.

Now, suppose you and your neighbor specialize by each only producing the good for which you have a comparative advantage.

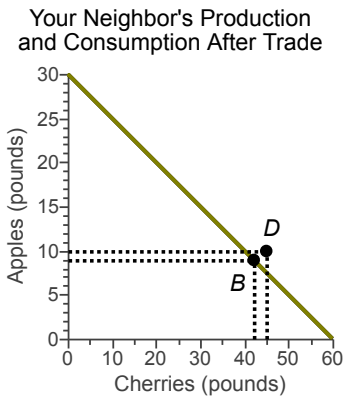
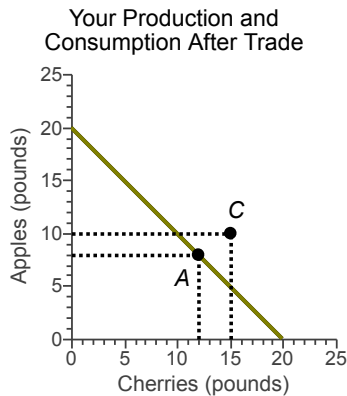
You give your neighbor half of your production for half of what he produces. (Enter all responses as integers.)

If you trade with your neighbor, then you will have additional coconut(s) after the trade and additional banana(s).

At the same time, your neighbor will be able to consume additional banana(s) and will be better off as a result of trade.



16.



Your neighbor's consumption point after trade is illustrated by point . (Enter your response as a letter.)

17. One of the great benefits of trade is

- ☐ A. that it makes it possible for society to become better off by increasing its production but not its consumption.
- ☐ B. that it makes it possible for society to become better off if individuals, firms, and countries specialize in producing goods and services for which they have an absolute advantage.
- ☐ C. that it provides society what it desires most in life even if it does not enhance the means for achieving its goals.
- ☒ D. that it makes it possible for society to become better off by increasing its consumption.

18. In economics, the term capital refers to

- ☐ A. the difference between a firm's assets and its liabilities.
- ☐ B. funds, including cash, used by businesses to hire resources.
- ☒ C. physical capital, such as machinery, that is used to produce other goods.
- ☐ D. the process of raising funds through the sale of stocks and bonds.

19. Which of the following are examples of intellectual property? *(Check all that apply.)*

- ☒ A. Books and films
- ☐ B. Pure air and water
- ☐ C. All non-natural resources
- ☒ D. Software
- ☒ E. Ideas for new products or new production techniques

The government seeks to protect intellectual property rights by granting

- ☐ A. copyrights.
 - ☐ B. patents.
 - ☐ C. franchises.
 - ☐ D. All of the above.
 - ☒ E. A and B only.
-

20. Suppose we can divide all the goods produced by an economy into two types: consumption goods and capital goods. Capital goods, such as machinery, equipment, and computers, are goods used to produce other goods.

Is it likely that the production possibilities frontier⁷ in this situation would be a straight line: ⁸ *click on the icon for an example* or bowed out: ⁹ *click on the icon for an example*?

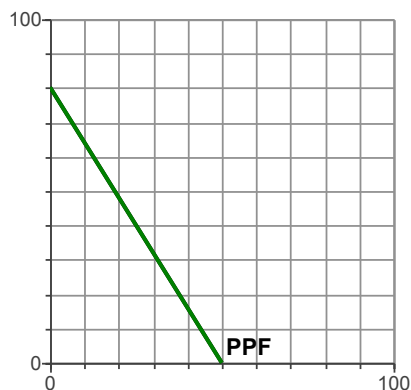
- ☒ **A.** The production possibilities frontier would likely be bowed out because not all resources are equally well suited to produce both consumption and capital goods.
- ☐ **B.** The production possibilities frontier would likely be a straight line because not all resources are equally well suited to produce both consumption and capital goods.
- ☐ **C.** The production possibilities frontier would likely be bowed out because resources are likely equally well suited to produce both consumption and capital goods.
- ☐ **D.** The production possibilities frontier would likely be a straight line because resources are likely equally well suited to produce both consumption and capital goods.

7: Definition

Production possibilities frontier (PPF) A curve showing the maximum attainable combinations of two products that may be produced with available resources and current technology.

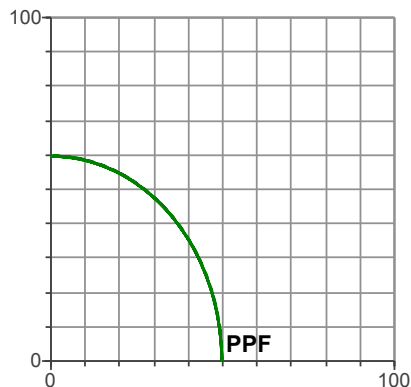
8: Graph/Chart

Production Possibilities: constant costs



9: Graph/Chart

Production Possibilities: increasing costs



21.

Use the following production possibilities frontier for a country to answer the following questions.

Which point(s) are unattainable? Briefly explain why.

- ☐ Point Z because it is inside the production possibilities frontier.
- ☐ None of the points because they all are feasible.
- ☐ Points W, X, and Y because they are on the production possibilities frontier.
- ☐ All the points because the production of each has an opportunity cost.
- ☒ Point V because it is outside the production possibilities frontier.

Which point(s) are efficient? Briefly explain why.

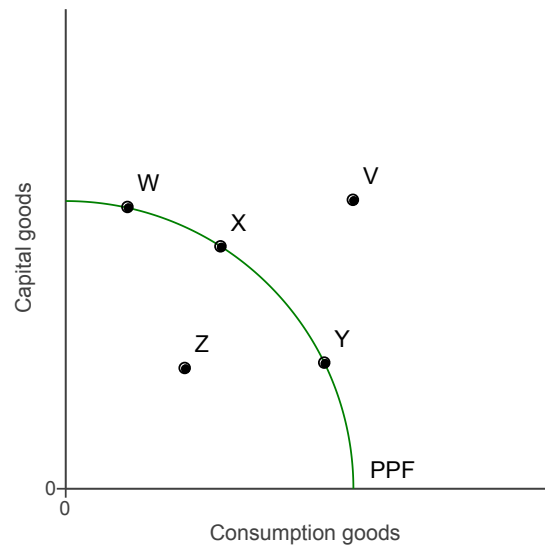
- ☐ Point W because it is where the most resources are used to produce capital goods.
- ☒ Points W, X, and Y because this is where maximum output is produced with available resources.
- ☐ Point V because it is where the most capital and consumption goods combined are produced.
- ☐ Points Z, W, X, and Y because they are attainable.
- ☐ Point Z because it is inside the production possibilities frontier.

Which point(s) are inefficient? Briefly explain why.

- ☒ A. Point Z because production there is not using all available resources.
- ☐ B. Points W, X, Y, and V because they are unattainable.
- ☐ C. Point V because it is unattainable.
- ☐ D. Points Z, W, X, and Y because they are attainable.
- ☐ E. Points W, X, and Y because they are on the production possibilities frontier.

At which point is the country's future growth rate likely to be the highest? Briefly explain why.

- ☐ A. Point X because it is an efficient point where a mix of capital and consumption goods are produced.
- ☐ B. Point V because it is where the production possibilities frontier will shift with growth.
- ☐ C. Point Z because it is where resources are saved for investments.
- ☐ D. Point Y because it is where the most resources are used to produce consumption goods.
- ☒ E. Point W because it is where the most resources are used to produce capital goods.



22.

Using the same amount of resources, the United States and Canada can both produce lumberjack shirts and lumberjack boots, as shown in the production possibilities frontiers in the figure to the right.

The United States has a comparative advantage in producing lumberjack boots.

Canada has a comparative advantage in producing lumberjack shirts.

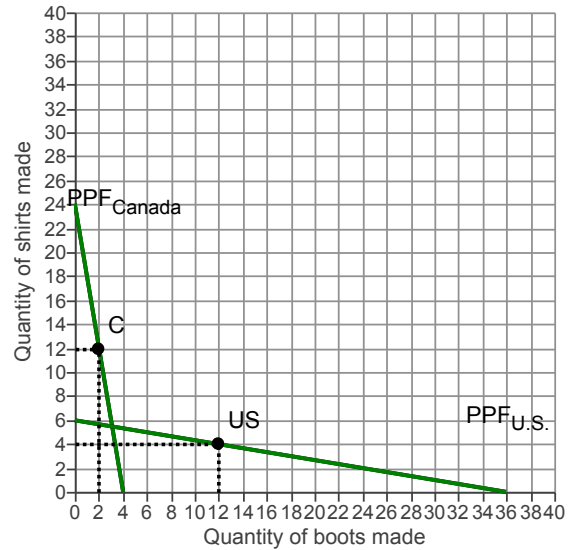
Does either country have an absolute advantage in producing both goods?

- ☒ A. Neither country has an absolute advantage in both goods because the United States can produce more boots but Canada can produce more shirts.
- ☐ B. Both countries have an absolute advantage in both goods because they both produce boots and shirts.
- ☐ C. Neither country has an absolute advantage in both goods because Canada can produce more boots but the United States can produce more shirts.
- ☐ D. Canada has an absolute advantage in both goods because it can produce more boots and more shirts.
- ☐ E. The United States has an absolute advantage in both goods because it can produce more boots and more shirts.

Suppose initially that the United States is consuming 12 boots and 4 shirts and Canada is consuming 2 boots and 12 shirts, as indicated in the figure. Then, suppose the United States and Canada specialize by each only producing the good for which they have a comparative advantage and then trade. In particular, suppose the United States trades Canada half of its production for half of what Canada produces.

The United States will have additional shirt(s) after the trade (enter a numeric response using an integer) and additional boot(s).

At the same time, Canada will be able to consume additional shirt(s) as a result of the trade and additional boot(s).



23. Suppose Iran and Iraq both produce oil and olive oil, which sell for the same prices in both countries. The following table shows combinations of both goods that each country can produce in a day, measured in thousands of barrels.

Iraq		Iran	
Oil	Olive Oil	Oil	Olive Oil
0	4	0	12
1	3	3	9
2	2	6	6
3	1	9	3
4	0	12	0

a. Who has the comparative advantage¹⁰ in producing oil?

- ☒ A. Neither country has a comparative advantage producing oil because their opportunity costs of producing oil are equal.
- ☐ B. Iran has a comparative advantage producing oil because its opportunity cost of producing oil is lower.
- ☐ C. Iraq has a comparative advantage producing oil because it can produce more oil.
- ☐ D. Iran has a comparative advantage producing oil because it can produce more oil.
- ☐ E. Iraq has a comparative advantage producing oil because its opportunity cost of producing oil is lower.

b. Can these two countries gain from trading oil and olive oil?

- ☐ A. These countries can gain from trade because Iraq has an absolute advantage producing olive oil.
- ☐ B. These countries can gain from trade because their opportunity costs of producing each good are the same.
- ☐ C. These countries can gain from trade because Iran has an absolute advantage producing oil.
- ☒ D. These countries cannot gain from trade because neither has a comparative advantage producing either good.
- ☐ E. These countries cannot gain from trade because neither has an absolute advantage producing either good.

10: More Info

Comparative advantage is the ability of an individual, a firm, or a country to produce a good or service at a lower opportunity cost than competitors.

24. In 2018, Canada joined 10 other countries in the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). The agreement will significantly reduce barriers to trade¹¹ among member countries (Australia, Brunei Darussalam, Canada, Chile, Japan, Malaysia, Mexico, New Zealand, and Vietnam). Some opponents of the original deal (the Trans-Pacific Partnership reached in 2015, which also included the United States) and the CPTPP argue that there is no way for countries as different as Canada and Vietnam to benefit from the trade agreement at the same time because Canada is so much more productive than Vietnam. Is there any way for Canada to gain from trading with a country over which we have an absolute advantage¹² in virtually everything? Is there a way for Vietnam to benefit as well?

Canada could benefit from importing products for which Vietnam has a comparative advantage . Vietnam could benefit from importing products for which Canada has a comparative advantage .

11: Definition

Trade is the act of buying and selling.

12: Definition

Absolute advantage is the ability of an individual, a firm, or a country to produce more of a good or service than potential trading partners, using the same amount of resources.